Cullen Kelley

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Objective

Detail-oriented cybersecurity professional seeking analyst role to leverage strong foundations in security operations, malware research, and vulnerability assessment to enhance organizational security posture and risk management objectives.

Education

Masters of Applied IT, Cyber Security Concentration Aug. 2024 – Dec. 2025
George Mason University — GPA: 3.82

Bachelor of Applied Science, Cyber Security Concentration Aug. 2021 – Aug. 2024
George Mason University — GPA: 3.83

- Phi Kappa Phi Honor Society Member, Dean's List (2022-2024)

Professional Experience

Software Security Consultant Jun. 2024 – Aug. 2024
Affinity DB — Leesburg, VA

- Conducted quality and security assurance audit of numerous Python and VBA applications written by third party contractors prior to production integration
- Created application documentation and a writeup documenting the review process and sharing findings with stakeholders
- Audited using common industry tools including Ghidra, x64dbg, oletools, and DidierStevensSuite MS tooling

eCommerce Department Manager and Trainer Sep. 2020 – Jun. 2024
Safeway — Great Falls, VA

- Promoted from entry position eCommerce shopper within six months of hire due to exceptional performance
- Identified vulnerabilities in multiple systems and responsibly disclosed them to Safeway IT security team
- Led operations of a three man team ensuring timely delivery of online orders with 99.3% on-timeness
- Launched eCommerce departments across Washington metropolitan area
- Independently produced training materials for on-boarding employees

Research & Technical Projects

Self-Driving Car BA Algorithm Research | *C++, Python, Google Ceres Solver* Sep. 2025 – Dec. 2025

- Assessed bundle adjustment performance on Oxford RobotCar, resulting in root mean square error of 1.871 (LM) and 1.701 (Dogleg) and final cost from 55,897.7 to 46,164.6
- Benchmarked solver efficiency: Dogleg cut BA time from 128.45s (LM) to 65.42s and iterations from 353 to 167

Biometric Authentication Modalities Research | *IREX, FIPS 201-3, SP 800-76-2* Jan. 2025 – May 2025

- Conducted research on biometric authentication modalities including fingerprint, iris, and facial recognition
- Analyzed tradeoffs in performance, security, and usability of each modality

GhostRace CVE Research | *C, Ghidra, x64dbg* Jan. 2024 – May 2024

- Researched and developed a Proof-of-concept exploit for CVE-2024-2193 (GhostRace)
- Analyzed previous branch-misprediction and speculative execution CPU vulnerabilities Spectre and Meltdown

Machine Learning Malware Detector | *Python, Rust, MariaDB, Google Cloud* Aug. 2024 – Present

- Developed CI/CD pipeline for ingesting malware samples for supervised machine learning malware detection
- Ingested 201,549 Windows malware samples for training data using label encoder and padding each entry
- Achieved a malware detection accuracy of 94% with k-folded cross validation, comparing to train-test split baseline

Cloud Service Model Risk Assessment | *Google Cloud, AWS* Aug. 2024 – Dec. 2024

- Conducted research on induced security risks inherent to SaaS, PaaS, and IaaS service models
- Applied industry research in addition to NIST (SP) 800-210

Cybersecurity Lab | *NixOS, Proxmox, OPNsense, Suricata, Wazuh, PLG, Tailscale, Keycloak* Feb. 2023 – Present

- Maintain a declarative IAC NixOS monorepo (flake) as single source of truth
- SOC-style detection stack: OPNsense/Suricata NIPS at perimeter, Wazuh SIEM on all hosts, Grafana Alloy feeding Prometheus/Loki
- Self-hosted Git forge with CI pipelines for supply-chain checks (Trivy SBOM, nix audit); secrets via sops-nix/Age
- Zero-trust access with Tailscale and Keycloak OIDC/MFA; used for HackTheBox labs and malware reverse engineering

Technical Skills / Accreditation / Awards

Certifications: Google Cybersecurity Professional Certificate, Security+ (exam scheduled for Aug.)
Languages: C, GoLang, Python, Bash, SQL
Security Tools: OPNsense, Suricata, Wuzuh, Shuffle, Nessus, nMap, Ghidra, FireEye FLOSS, x64bdg, Autopsy, FTK Imager
Additional Awards: Phi Kappa Phi Honors, Eagle Scout, Scouting Heroism Award, Sons of the American Revolution member